

A RAG System for Advertising Legality Detection Based on Taiwanese Regulations

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Abstract

This project aims to evaluate the legality of advertising content under Taiwanese law. By integrating semantic retrieval and language model reasoning, we develop a Retrieval-Augmented Generation (RAG) pipeline that takes user-inputted advertising statements and assesses potential legal risks based on relevant case data and regulatory texts.

1 Screenshot of the final result

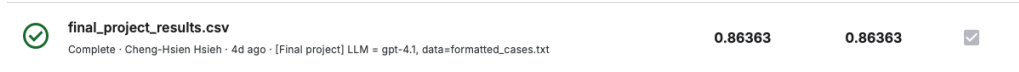


Figure 1: Screenshot of the final result

2 Methodologies

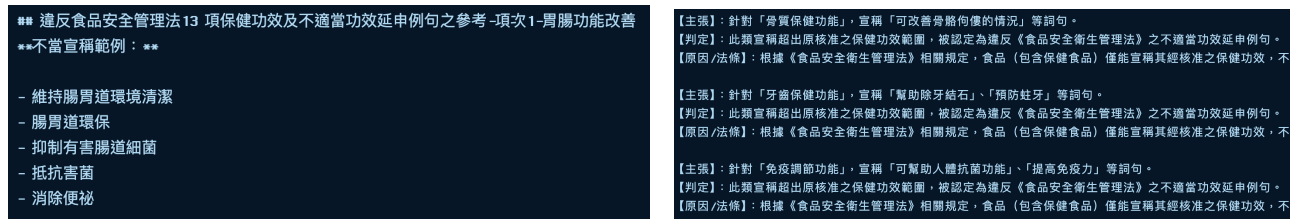
Our methodology consists of the following components:

2.1 Data Pre-processing

First of all, we are provided with some data about Taiwanese regulations and relevant cases. To utilize the precious information, we have to put it in our **prompt** in certain forms to make the Judge Agent perform better.

However, the structure of the data is somewhat complex and inconsistent so that we cannot easily access the information. To deal with this, we have implemented **two version** of pre-processing to organize our data into easily accessible format:

1. **Markdown Format:** In this version, we implemented several **organizer** programs to help us extract the information in each JSON data file and organize it into simple and concise markdown paragraphs.
2. **AI Helper:** In this version, we employed a **LLM (gemini-2.5-flash-preview-05-20)** to assist us in reading each individual JSON data file and organizing it into a highly structured and minimal format.



(a) Markdown format

(b) AI organized format

Figure 2: Comparison between Markdown and AI organized format

After consistently organizing the data, we could easily and clearly access the appropriate information.

2.2 Prompt Design

Prompt serves as the core component in our project. In fact, the core technique of the entire RAG system is to capture key information and put it into our prompt. And prompt is the only thing that will come into contact with Judge Agent at the end.

Based on the `baseline_prompt`, we added two parts in our prompt: **Regulations** and **Similar cases**.

- **Regulations:** Since the regulations are not so many and should serve as the principals of judging, we collect the related laws and add them to the prompt.
- **Similar cases:** In addition to regulations, we should also provide some judged cases that are similar to the input advertisements to help Judge Agent make a better judgement.

2.3 Embedding & Retrieving

However, since there are so many cases, obtaining similar cases becomes quite difficult. Thus, we apply the core method of a RAG system: **Embedding** and **Retrieving**.

We embedded both the cases (embed each markdown paragraph or AI organized paragraph) and input queries into vector space. And FAISS is used to create an efficient similarity index for retrieval. For any given advertising sentence, the system retrieves the top- k most semantically relevant legal cases by computing cosine similarity between the query and all indexed cases.

Note: If the embedding input exceeds the model's token limit, we apply truncation to preserve only the first 8192 (max tokens) tokens of the text.

2.4 Violation Judgment

We use `gpt-4.1` as Judge Agent for illegality assessment. The prompt includes:

1. Role and task description
2. Compliance judgment guidelines
3. A list of permissible advertising phrases
4. Step-by-step analysis framework
5. Summarized legal provisions (only the parts defining illegality, excluding penalty clauses)
6. Case examples retrieved by the RAG model (3 legal and 3 illegal)
7. The advertisement content to be judged

The model is instructed to determine legality, provide a probability of illegality (0–100%), reasons, **violated legal provisions, and citations to relevant cases and laws** (the **Bonus** part).

We previously attempted to avoid having the model output a raw probability score due to high variability and lack of sensitivity between close values like 70% and 90%. Instead, we tried a five-level rating system (1 to 5) to represent risk levels from “completely legal” to “completely illegal,” hoping the model would focus on relative severity rather than precise numbers.

However, this approach resulted in outputs clustering around level 4, reducing discrimination power between clearly legal and clearly illegal cases.

Ultimately, we reverted to using probability-based assessment as the main method.

Response:

違法機率：10%

相關法條內容及解釋：

本則廣告內容聚焦於：「五大黃金順暢國際配方,保護消化道全機能! 益生菌X益生元X後生元,改變細菌生態、共同養護消化道健康,更加了消化酵素X蔬果發酵精華,提升吸收滋養消化機能。」

其主要描述為：「保護消化道全機能」、「改變細菌生態」、「提升吸收滋養消化機能」等，並未有醫療效能之用語，例如治療疾病、治癒、改善特定病理狀況等。

上述語句屬於目前衛福部公告允許用於食品/健康食品的常用表述，諸如「改變細菌叢生態」、「保護消化道」、「提升吸收滋養消化機能」，皆屬常見許可詞語，未涉過度誇大療效或明示醫療效能。

相關法條如下：

- 依據《食品安全衛生管理法》第28條第1項規定，食品廣告不得涉及醫療效能。
- 依據《健康食品管理法》第14條，健康食品不得涉及醫療效能之內容。
- 但根據合規性判斷原則及比對公告允許表述，本案僅宣稱與消化道機能、生理調節、消化吸收等有關，仍屬合法範圍。

因此，違法機率偏低。若未有超出「改變細菌叢生態」、「提升吸收滋養消化機能」、「保護消化道」等公告容許詞句範圍，推定其合規。

Figure 3: Response

2.5 Postprocessing

From the Judge Agent’s response, we extract the numerical probability of illegality and the reasoning. If the probability exceeds the threshold, the advertisement is deemed illegal.

Through extensive testing, we found that a threshold of 80% yields a stable and discriminative performance, with an F1 score around 0.85. Further ensemble voting on high-performing predictions raised the final F1 score to 0.86.

3 Discussion

3.1 Strategy for Using Laws and Cases

It is worth noting that in our current implementation, only the *cases* are embedded and retrieved, while the *regulations* are all directly inserted into the prompt. This design decision is based on two main considerations.

First, the total amount of regulations is **relatively small** compared to the large volume of case data. This makes it feasible to include the full set of relevant regulations in the prompt without exceeding the token limit. And as mentioned above, regulations are suitable as the concept of principal to provide to Judge Agent, so it would be a better choice to put all of them into prompt.

Second, there is a stylistic gap between **legal language** (formal and precise) and **query language** (casual and marketing-oriented). This mismatch makes it difficult for vector-based RAG models to effectively retrieve the correct legal sections, thus degrading retrieval quality and reasoning accuracy.

In contrast, **cases** contain richer and more concrete contextual language, the text in regulations typically lacks sufficient surface-level semantic similarity to advertisements. As a result, embedding and retrieving regulations may not effectively capture the intended legal relevance.

Therefore, we believe that directly including the regulations in the prompt leads to better performance and more faithful legal interpretation in the generation stage.

3.2 Reasoning Process

In this section, we present the results of our empirical tests focusing on the prompt design, particularly the instruction format. Initially, our prompt only included the relevant legal regulations and asked Judge Agent to judge whether the advertising statement was legal or not. This setup resulted in a relatively low F1-score of **0.60869**.

However, by incorporating a structured **reasoning process** into the instruction, the F1-score significantly improved to **0.82539**. The improved instruction included the following legal reasoning steps:

Compliance Judgment Guidelines:

```
### 合規性判斷
- **無罪判定原則**：未找到確切違反法律的證據時，應推定其為無罪
- **從寬認定合法性**：不捏造或過度解讀廣告文字，例如推論產品「暗示」了哪些效用，應減少此類較無證據之推論，但如果是**藥物**宣稱**
- **比對允許使用的廣告用詞**：
  - 「調整體質」、「促進新陳代謝」、「保護消化道全機能」、「改變細菌叢生態」、「在嚴謹的營養均衡與熱量控制，以及適當的運動條件下，適量
  - 「能完整補充人體營養」、「提升、調節生理機能」、「幫助入睡」、「青春美麗」、「排便超有感」、「給你排便順暢新體驗」、「改變細菌叢生態」
```

Figure 4: Compliance judgement guideline

Step-by-Step Analysis Framework:

```
## 分析步驟
1. **解析廣告內容**：先觀察分類廣告商品屬於哪類產品，接著檢視其是否涉及療效誇大。
2. **案例文件與法規比對**：根據以下法律文件與案例，提供比對結果。
3. **判斷違法機率**：依據法律及案例進行風險評估。
```

Figure 5: Step-by-step analysis framework

This structured reasoning not only improves Judge Agent understanding but also helps align the decision-making process with real-world legal practices.

 final_project_results.csv Complete · Cheng-Hsien Hsieh · 9d ago · [Final project] Only system prompt with laws' description	0.60869	 final_project_results.csv Complete · Cheng-Hsien Hsieh · 9d ago · [Final project] Add baseline prompt, (cases) embedding and retrieval.	0.82539
(a) Without reasoning process		(b) With reasoning process	

Figure 6: Comparison of w/wo reasoning process

3.3 Necessity of the RAG Model

During experiment, we also wondered that whether the Judge Agent could achieve high performance using only the regulations, without retrieving similar cases. To evaluate the necessity of adopting a RAG-based approach, we conducted an experiment in which the retrieval module was removed.

The result shows that while the model still performs reasonably well relying solely on the legal text, there is a noticeable drop in performance. Specifically, the F1-score decreased from **0.83720** (with RAG) to **0.80645** (without RAG).

This suggests that the inclusion of retrieved, semantically relevant legal cases provides valuable context and enhances the model's ability to make more accurate and reliable legal assessments. We believe that this improvement stems from the fact that real cases often offer implicit patterns, practical interpretations, and nuanced reasoning that may not be directly captured by legal texts alone. While regulations provide the normative basis, case examples ground these norms in real-world applications, offering the model concrete parallels for better generalization.

In other words, the retrieval step not only supplements the prompt with additional domain knowledge, but also bridges the semantic gap between abstract legal terms and natural advertising language, which is essential for determining borderline legality in complex or ambiguous scenarios.

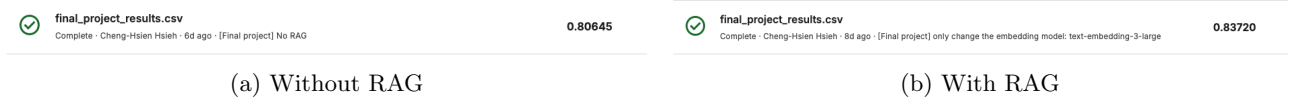


Figure 7: Comparison of w/wo the RAG system

3.4 Different RAG Frameworks

In our pursuit of enhanced performance, we also explored alternative RAG frameworks to determine if they could improve the accuracy of our legality assessment. We specifically selected to use a Node-based RAG - **NodeRAG** [1] system. For this experiment, we utilized OpenAI’s **GPT-4o-mini** as the large language model and **text-embedding-3-large** for generating embeddings.

This advanced approach moves beyond simple document chunking by dissecting source materials into a structured knowledge graph of interconnected “nodes.” The retrieval information for each node is meticulously organized into several fields. For instance, the retrieved data for a node might include: **high_level_element** (representing the parent document or a major section), **attribute** (describing the node’s function, e.g., ‘ruling’ or ‘reasoning’), the core **text** of the **semantic_unit**, and its **relationship** to other nodes.

This structure allows for a more sophisticated retrieval process. Instead of fetching isolated text blocks, the system targets the most relevant child nodes (the granular ‘semantic_unit’) and, by leveraging the ‘relationship’ data, reconstructs their context by including parent nodes (‘high_level_element’) or relevant sibling nodes. For example, when retrieving a specific legal argument, the system also provides the overarching case summary it belongs to. This strategy is designed to provide richer, more focused, and structurally aware context to the LLM than retrieving entire, undifferentiated document chunks.

Despite its theoretical advantages in providing more precise, structured context, our experiment with the Node-based RAG yielded an F1-score of **0.81944**. This result did not represent a substantial improvement over our refined prompt-based model. We hypothesize that while providing richer context is beneficial, an excessive amount of interconnected information, as supplied by the NodeRAG, may have inadvertently confused the LLM, preventing it from effectively distinguishing the most critical information and leading to a lack of performance improvement. The marginal gain suggests that for our specific task, the added complexity of the hierarchical retrieval system did not translate into a significant performance boost. This outcome reinforces our conclusion that meticulous **prompt engineering**, including the integration of structured reasoning guidelines, remains the more impactful and effective strategy for optimizing the model’s performance.

4 Improvement & Future Outlook

4.1 Model Selection

Due to limitations in local computational resources and environment configuration, we opted to utilize cloud-based API models rather than running models locally. This approach enabled us to access more powerful and up-to-date LLMs without the burden of managing local infrastructure.

Throughout the development process, we experimented with multiple model configurations to enhance system performance. For the embedding stage, we initially adopted **intfloat/multilingual-e5-large**, but later switched to OpenAI’s **text-embedding-3-large**, which provided better semantic alignment and improved retrieval quality.

For the final legal judgment agent, we started with **gemini-2.5-pro-preview-05-20**, and later transitioned to **gpt-4.1**. These model upgrades contributed to a noticeable improvement in the overall system performance.

4.2 Query Preprocessing

Since the queries are sourced from web crawlers, they are highly unstructured and noisy. They often contain irrelevant information like platform names, price tags, promotions, or even non-advertising text. This noise confuses the Judge Agent and reduces retrieval precision, possibly leading to semantically irrelevant case retrieval. Moreover, such noise causes significant token waste, increasing cost.

While we use `gpt-4o` as Parser Agent to extract advertising claims from raw queries, this step is computationally expensive. Even if performed only once, it may incur substantial token and resource costs in production.

More fundamentally, this problem should be addressed earlier. If the crawler can pre-filter for actual advertising blocks or extract product slogans and descriptions, it will significantly reduce the burden on subsequent stages.

We manually inspected some poorly-formed queries, finding many with irrelevant or incomprehensible content.

```
93,"3party/jqm-datebox-1.4.5.min.css,3party/selectize/selectize.default.css,3party/owl.carousel/owl.carousel.min.css,3party/shhos/shhos.css,3party/jquery.fancybox.min.css,3party/animate.min.css,index.css,hover.css,base.css,element.css,style.css,landingpage.css,store.css,form-style.css,layout.css,status.css,event.css,user.css,register.css,start_event.css,mobile_tour.css,mobile_guideline.css,create-event.css,find-event.css,fontello.css,2017-message.css,search.css,regular-header.css,profile.css,label.css,banner.css,index-phone.css,label-page.css,new-label-page.css,rwd-product-page.css,3party/jquery-1.11.1.min.js,3party/setQueryMobile.js,3party/jquery.mobile-1.4.5.min.js,3party/jquery.scrollTo.min.js,3party/jqm-datebox-1.4.5.min.js,3party/jsonfn.min.js,3party/jquery.cookie.min.js,3party/jquery.blockUI-2.70.0.js,3party/selectize/sifter.js,3party/selectize/micrologin.js,3party/selectize/selectize.min.js,3party/owl.carousel.min.js,3party/exif.js,3party/iscroll-zoom.js,3party/bootstrap/js/bootstrap.min.js,3party/masterslider/masterslider.min.js,3party/URI.js,3party/jquery.fancybox.min.js,3party/jquery.timers.js,util/mobile.js,social/social.js,util/googleAnalyticsEvent.js,util/android.js,util/areas.js,util/category.js,tab.js,shhos/shhos.js,slick/slick.js,util/imageUploader.js,system/setUserType.js,system/webPush.js,util/cookieUtils.js,util/webStorageUtils_v3.js,util/determineUri.js,util/shoppingDialogUtil.js,util/mobileMailUtil.js,util/mobileMessageUtil.js,messages/receiverPrepare.js,stores/indexStores.js,index/indexIndex.js,index/indexMap.js,index/indexSearch.js,index/indexCategoryMenu.js,header.js,menu.js,events/join/index.js,events/join/pay/prepare.js,events/join/description.js,events/modify/prepare.js,events/join/insertCustomItem.js,events/modify/index2016.js,events/prepare/prepareProducts.js,events/prepare/prepareFreeProducts.js,events/prepare/prepareEventInfo.js,events/prepare/prepareHostInfo.js,events/prepare/complete.js,fast/indexFast.js,fast/indexCategories.js,fast/indexCampaignTags.js,fast/indexLabels.js,fast/fastGroupBuy.js,fast/deliverDirectoryList.js,fast/deliverDirectoryModify.js,fast/listFastPlaces.js,fast/fastDirectBuy.js,fast/enrollPay.js,fast/deliverPay.js,fast/fastProductInfo.js,fast/fastDirectBuyInfoStepOne.js,fast/viewedProduct.js,store/fastStoreIndex.js,user/malls/index.js,user/malls/detail/eventDetail.js,user/malls/message/eventMessage.js,user/malls/inbox/index.js,user/malls/detail/defaultDetail.js,malls/writePrepare.js,user/malls/detail/meetingCheck.js,user/comment/index.js,user/profile/index.js,user/profile/myCoupon.js,deliver/tools.js,util/theropayment.js,user/malls/detail/deliverDetail.js,user/malls/message/deliverMessage.js,store/index.js,store/content.js,store/deliverAdd.js,store/storeEvents.js,status/index.js,point/insertPoint.js,bonus/bonusUtils.js,status/listSuggestFriends/index.js,favorites/indexFavorites.js,util/youtubeVideo.js
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【台灣生薑】樂齡養生薑(30包入/盒) 30包入/盒 | iherno聯合購 APP下載 與我合作 新手上路 回首頁 限時大舍贈 名店街 找舍贈 破萬件商品 天天免費 搜尋 0 購物車 登入 免費註冊 折扣優惠 購買方案 特色介紹 商品說明 免運購買 加入購物車 開團 搜尋紀錄 熱門關鍵字 今天有面交! 送上X張折價券 請享用! 2/17前購買肉捲加送2捲 大成花椰菜米 5種口味任選 智慧體重計可記錄15項數據 加倍燃燒 運動不怕操過期 Y 保健藥用 8et; 保健食品 8et; 益生菌 免運 享優惠 【台灣生薑】樂齡養生薑(30包入/盒) 30包入/盒 付款後最慢3個工作天內出貨 特價6900 | 83折後 \$573 共1盒 領取成功, 滿額結構使用可折200元! 結構輸入過年肥田說辦 滿1800折 8et; \$200 領取折價券 結構輸入 0210限定下殺 83折 19496740 2025/02/10 11:05:42 折扣倒數 0 天 00 時 00 分 00.8et; 0 秒 領取折價券 登入查看所有折價券 已售出17份 下單即享禮品 紅利回饋1% 紅利回饋1% 美安回饋3% shpack回饋 下單紅利折抵可低至 73 折 快速出貨 免運購買 加入購物車 你可能也想看 【乾鹿軒活瓷】日出之地遠紅外線毛毯 附精美禮盒 原價\$3880 \$3298 起 再9折 \$2968.2 起 8et; 【I LEKE 艾樂可】樂舒適草蓆每膠套 原價\$33.3 \$719.1 起 再89折 \$640 起 8et; 【大漢酵素】淨暢酵素錠 原價\$1020 \$865 起 再9折 \$839 起 售出16份 8et; 【詠統】董董好黑蒜精 原價\$200 \$279.2 起 再93折 \$259.7 起 8et; 【詠統】黑蒜膠囊PLUS 原價\$44.3 \$1050 起 再9折 \$945 起 8et; 三立電視-天之驕女歐連纖膠囊買6盒送4盒共10盒再贈送24K黃金電動按摩T字棒 原價\$232.7 \$1030 起 再86折 \$885.8 起 8et; 【自在食刻】低卡蒟蒻飯+罐頭套餐 原價\$1040 \$799 起 再9折 \$719.1 起 售出18份 8et; 【河洛嚴選】特選番紅花絲(小包裝) 1g 原價\$399 \$452.4 起 再93折 \$420.7 起 8et; 【女神伊登】-水嫩光潤玻尿酸 多組優惠 原價\$56 \$650 起 再89折 \$578.5 起 8et; 【益源泉】羊肚菌菇 50g/包 羊肚菌 特級菌菇 原價\$57 \$799 起 再88折 \$703.1 起 8et; 【einfitt】純素米豆腐白粉 (10包) 原價\$65 \$580 起 再94折 \$545.2 起 售出153份 8et; 【潮味館】炙燒豬排翅片 原價\$480 \$394.8 起 再86折 \$339.5 起 售出6份 8et; 【凍凍鮮】智利鮭魚非力 225g 原價\$350 \$255 起 再93折 \$237.2 起 8et; 【築宅】祖傳油二節翅(180g)/油雞爪(380g) 原價\$399 \$115 起 再93折 \$107 起 8et; 【福忠字號】吸吸香辣臭豆腐冬粉 原價\$268 \$223.1 起 再97折 \$216.4 起 8et; 【如意生技】純素香菇丸/海苔丸/紅毛苔丸任選 原價\$750 \$149 起 再94折 \$140.1 起 8et; 贈品資訊 - 贈品項任選 1 份 下單即享禮品 《台灣生薑》Dr's Formula控油抗屑洗髮精(升級版)三代100g - 100gx1瓶 配送與付款 配送資訊 可出貨日: 付款後最慢3個工作天內出貨 付款資訊 運費說明: 全館免運費 發票方式: 電子發票

Figure 8: JavaScript or unrelated product names

Even so, in most cases, Parser Agent can still extract usable advertising content.

Thus, for real-world deployment, we strongly recommend building a cleaner advertising dataset. This not only reduces preprocessing costs but also improves overall model reliability and judgment quality.

References

- [1] T. Xu, H. Zheng, C. Li, H. Chen, Y. Liu, R. Chen, and L. Sun, “Noderag: Structuring graph-based rag with heterogeneous nodes,” 2025. [Online]. Available: <https://arxiv.org/abs/2504.11544>